

Please note that phones and calculators are not allowed during the exam.

Exercise 1 (8.5 points). 1) Let n people place their phones on a table in a linear arrangement. Each phone has three possible modes: Silent (S), Ringing (R), and Vibrating (V). Determine the number of configurations for the phone modes under each of the following conditions:

- No restriction: Each phone can be in any of the 3 modes.
- Last 5 phones in the same mode (Assume $n \geq 5$.)
- Exactly r phones are Vibrating.
- No two consecutive phones have the same mode.
- There are exactly k_1 phones in Silent (S) mode, k_2 phones in Ringing (R) mode, and k_3 phones in Vibrating (V) mode, with $k_1 + k_2 + k_3 = n$.

1. Using a combinatorial argument, prove that $\sum_{k=0}^n \binom{n}{k} 2^k = 3^n$.

2. What is the coefficients of $x^{48}y^{105}z^{70}t^{51}$ in the expansion of $(x^4 - y^7 + 7z^2 + 4t^3)^{79}$?

Exercise 2 (6.5 points). A competition is being organized for n participants. Each participant is accompanied by three team member: a friend, a coach, and an assistant. This makes a total of $4n$ people. These $4n$ people need to be divided into groups of 4 for team challenges. Each group must have One participant, One coach, One friend, and One assistant, such that no participant can be in a group with all three of their own team members (their friend, coach, and assistant). In how many ways can these $4n$ people be divided into groups of 4 for the team challenges?

Exercise 3 (5 points). Determine whether each of the following statements is True or False. Justify your answer.

- In any simple graph, there must be at least two vertices with the same degree.
- Every subgraph of a graph is a vertex-induced subgraph.
- A trail in a graph is a walk that does repeat some edges.
- Every circuit in a graph is also a cycle.
- If a graph G is connected, then its complement must be disconnected.
- In a simple graph, if the neighbors of two distinct vertices are identical, then the graph must contain a triangle.
- Every connected graph with n vertices has a spanning subgraph that is a path with n vertices.